

JOHN JOSEPH ZAGARIAH DANIEL

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Data & analytics professional with 3+ years of experience turning operational and scientific data into dashboards, automated pipelines, and decision-support tools that leadership relies on. Completing a Master's in Computer Science (Applied AI) with hands-on work across data engineering, statistical analysis, BI reporting, and ML. Strong communicator who delivers in fast-paced, cross-functional, federal and enterprise environments.

TECHNICAL SKILLS

- **ML & AI:** PyTorch, HuggingFace Transformers, LoRA/PEFT, RAG, Agentic RAG, Scikit-learn, Optuna, LangChain, LangGraph, Sentence-Transformers
- **LLM & Agents:** Groq API, Google Gemini API, HuggingFace Inference API, Tool Calling, Function Calling, ReAct/Reflexion Agents, Guardrails, Multi-Agent Orchestration, Prompt Engineering, Synthetic Data Generation
- **Languages:** Python, Java, SQL, Shell, JavaScript
- **Frameworks & Tools:** FastAPI, Pydantic, Jina AI, Alembic, Next.js, Git, Splunk
- **Data & Infrastructure:** PostgreSQL, Redis, ChromaDB, MongoDB
- **Cloud & DevOps:** AWS (EC2, S3, RDS, CloudWatch), OpenTofu, Docker, Vercel, GitHub Actions (CI/CD), HuggingFace Inference Endpoints
- **Security & Quality:** JWT Auth, bcrypt, Sentry, Opik, Ruff, MyPy

EXPERIENCE

Agriculture and Agri-Food Canada (Co-Op)

Ottawa, Ontario, Canada

Machine Learning Research Associate – Bioinformatics

Jan 2025 - Dec 2025

- Built an end-to-end, config-driven genomic prediction pipeline as a modular Python package, replacing fragmented research workflows with a reproducible architecture and cutting manual setup time by ~50%.
- Integrated traditional statistical, machine learning, deep learning, and graph-based models behind a unified interface, with automated data validation, experiment tracking, and Optuna-based hyperparameter tuning to improve reproducibility and accelerate model selection.
- Led development of analytical visualizations including gene-density maps and genetic-distance analyses that enabled clearer statistical interpretation and directly informed downstream breeding decisions.
- Co-authored 2 research papers, translating complex model outputs and statistical findings into clear written analysis for technical and scientific stakeholders.

Accenture Solution Pvt. Ltd

Chennai, Tamil Nadu, India

Packaged App Development Analyst

Aug 2021 – Dec 2023

- Built interactive analytics dashboards in Splunk and Power BI to monitor cloud spend, VM utilization, database health, batch jobs, security posture, and API performance giving leadership centralized, real-time visibility for decision-making across multiple application teams and replacing fragmented manual reporting.
- Designed an Azure cost-intelligence data pipeline that ingested resource telemetry and billing data across subscriptions via the Cost Management API, Azure Monitor / Log Analytics, and Resource Graph queries, authenticated securely through Service Principals and Key Vault, and landed it in Data Lake / Blob Storage for downstream analysis.
- Orchestrated the pipeline with Azure Data Factory and Synapse, transforming raw telemetry into curated datasets that surfaced spending patterns, idle-resource waste, and optimization opportunities directly informing leadership budget and security decisions.
- Deployed serverless Azure Functions (Python) for scheduled health checks on URL availability and VM uptime, strengthening incident response and operational awareness across the estate.
- Designed and built an internal Operational Readiness Testing platform adopted by 3 application teams, replacing spreadsheet-driven workflows with Python automation, REST integrations, and custom Splunk commands; enabled one-click assignment of 200+ tickets per cycle and centralized leadership visibility into release readiness.
- Engineered data parsing and event-handling pipelines (props.conf / transforms.conf) for high-fidelity, CIM-compliant ingestion across UNIX, Linux, and Windows sources, and managed a distributed Splunk environment (indexers, search heads, forwarders) to ensure data availability and integrity.

PROJECTS

MedCliniQ: Benchmarking & Mitigating Hallucinations in Medical QA LLMs

Jan 2026 – Apr 2026

- Benchmarked 3 open-weight LLMs (BioMistral-7B, Gemma-7B-IT, Llama-3.1-8B) across 18 model-method configurations and 3,000 medical test questions spanning 4 standardized formats (T/F, MCQ, List, Short Answer).
- Systematically evaluated Prompt and Model Design strategies (Zero-Shot, Role-Prompting, RAG, RAG-CoV, LoRA fine-tuning, RAG+LoRA), demonstrating that LoRA consistently outperformed prompting-only approaches with F1 scores above 0.89 on binary verification tasks.
- Designed a dual evaluation framework combining F1-based scoring for closed-ended tasks and multi-level semantic similarity (word, sentence, paragraph) for open-ended clinical explanations.
- Deployed the top-performing model (Gemma-7B-IT + LoRA) as a full-stack medical chatbot using Streamlit, FastAPI, Firebase Auth, Firestore, and HuggingFace Inference Endpoints.

EDUCATION

University of Ottawa

Ontario, Canada

Master of Computer Science, Applied Artificial Intelligence

Jan 2024 – Apr 2026

Panimalar Engineering College, Anna University

Chennai, Tamil Nadu, India

Bachelor of Engineering, Computer Science and Engineering

Jun 2017 – Apr 2021